Applied Deep Learning

Convolutional Neural Networks

Last Week

Applying Activation Functions and Reducing Overfitting

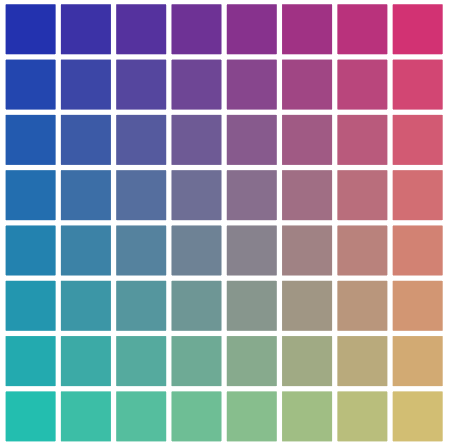
- What are the four activation function types we discussed?
- What is the problem with overfitting?
- What are the key tools to overcome overfitting?

This Week - CNNs

How Does a Computer Represent an Image?

- An image is a grid of tiny squares called pixels.
- Each pixel contains numerical information about brightness or colour.
- Image dimensions describe the number of pixels across its width and height.





Grayscale vs RGB Images

- Grayscale image: Width × Height × **1**
- RGB image: Width × Height × **3**
- The final dimension is the number of colour channels.



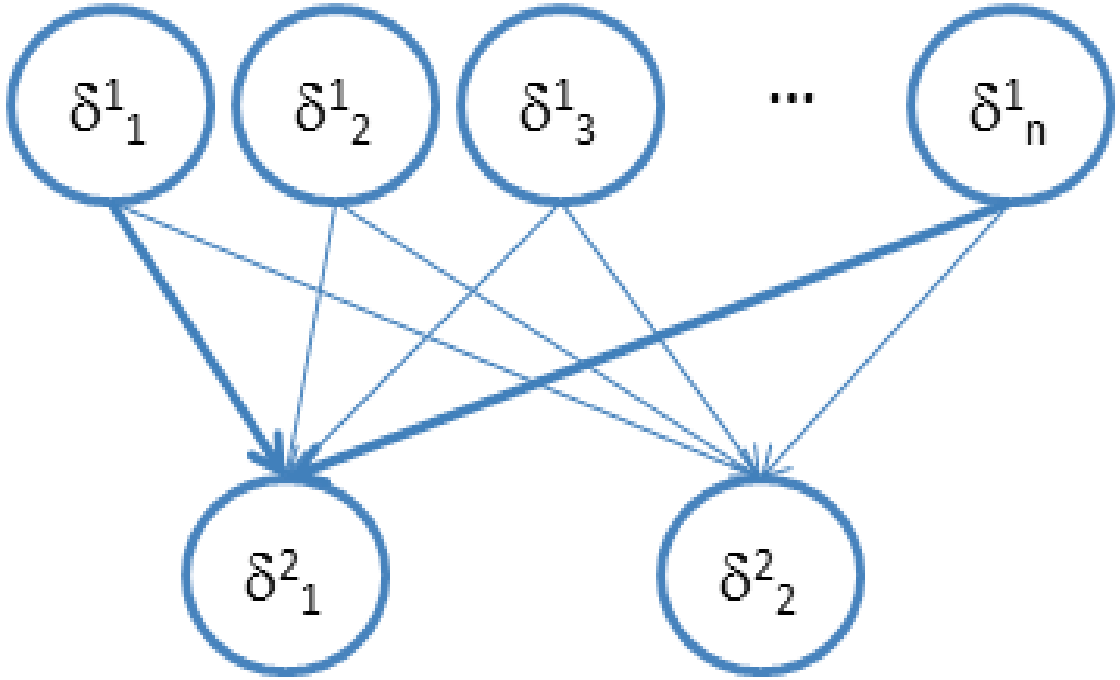


What are Hidden Layers Doing?

Learning Feature Detectors

Sub Question 1: What sort of features can the neurons in a first hidden layer learn in a fully connected network?

Sub Question 2: Could a Hidden Layer learn to recognize a specific black and white face in a specific position in an image?



Problem 1: Computational Explosion

- The number of weights to be learned in a fully connected layer between input units L1 and hidden units L2 is:

$$(size(L1) + 1) \times size(L2)$$

- How many weights (or parameters) are required for a 256 x 256 three colour image that is to be fully connected to a hidden layer with 128 neurons?
  - 256 x 256 x 3 = 196608
  - 196608 x 128 = 25+ million

25 Million!!!!

Problem 2: Local vs. Global Detection





Interesting things in images are usually **translation invariant**

In early layers we can focus on local features and leave global features until later

Edge Detection – Example of a Local Feature detector that can be applied to a whole image

An edge is a steep gradient in intensities

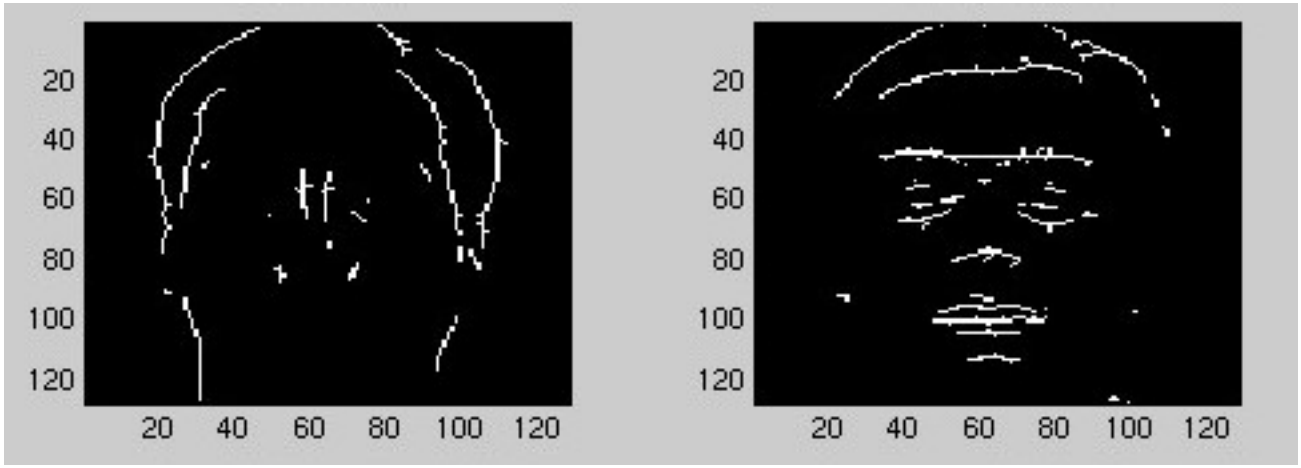
-1	0	+1
-1	0	+1
-1	0	+1

Gx

+1	+1	+1
0	0	0
-1	-1	-1

Gy

Weight matrix for edge detection



https://www.owlnet.rice.edu/~elec539/Projects97/morphjrks/moredge.html

Edge Detection – Prewitt Filter/Kernel

Image (3 × 3 pixels) Filter (3 × 3)



$$(-1\times100)+(0\times100)+(1\times100)$$
$$+(-1\times100)+(0\times100)+(1\times100)$$
$$+(-1\times100)+(0\times100)+(1\times100)$$

Response = -300 + 0 +300 = 0

Image (3 × 3 pixels) Filter (3 × 3)



$$(-1\times0)+(0\times0)+(1\times255)$$
$$+(-1\times0)+(0\times0)+(1\times255)$$
$$+(-1\times0)+(0\times0)+(1\times255)$$

Response = 0 + 0 + 765 = 765

Edge Detection – Prewitt Filter/Kernel

Image (3 × 3 pixels) Filter (3 × 3)



$$(-1\times255)+(0\times255)+(1\times0)$$
$$+(-1\times255)+(0\times255)+(1\times0)$$
$$+(-1\times255)+(0\times255)+(1\times0)$$

Response = -765 + 0 + 0 = -765

Image (3 × 3 pixels) Filter (3 × 3)



$$(-1\times100)+(0\times110)+(1\times120)$$
$$+(-1\times100)+(0\times110)+(1\times120)$$
$$+(-1\times100)+(0\times110)+(1\times120)$$

Response = -300 + 0 + 360 = 60

Edge Detection – Example of a Local Feature detector that can be applied to a whole image

An edge is a steep gradient in intensities

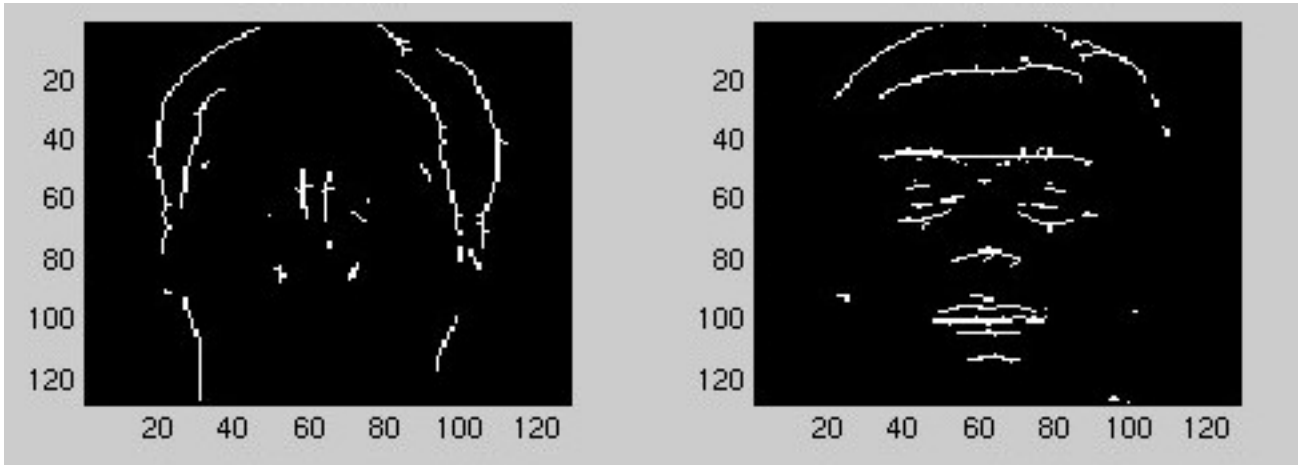
-1	0	+1
-1	0	+1
-1	0	+1

Gx

+1	+1	+1
0	0	0
-1	-1	-1

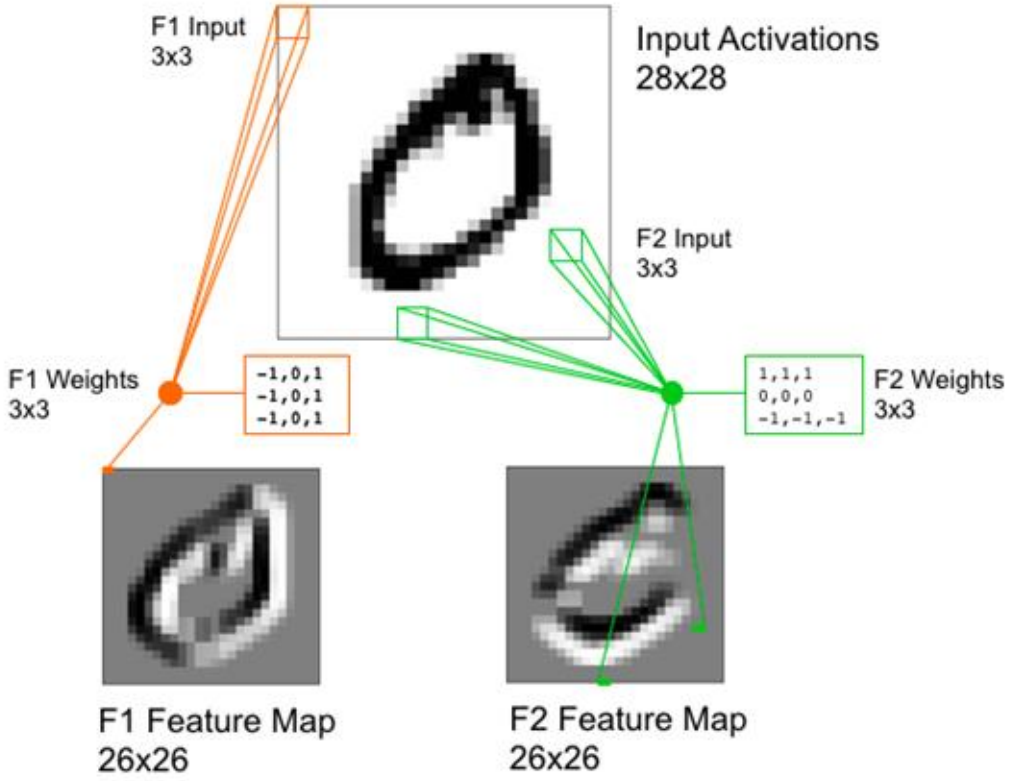
Gy

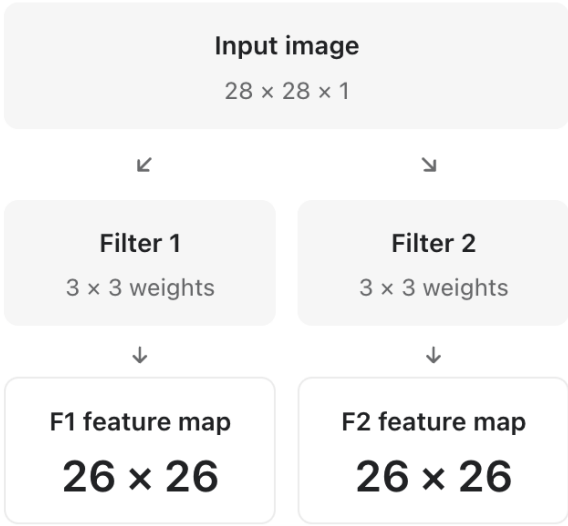
Weight matrix for edge detection



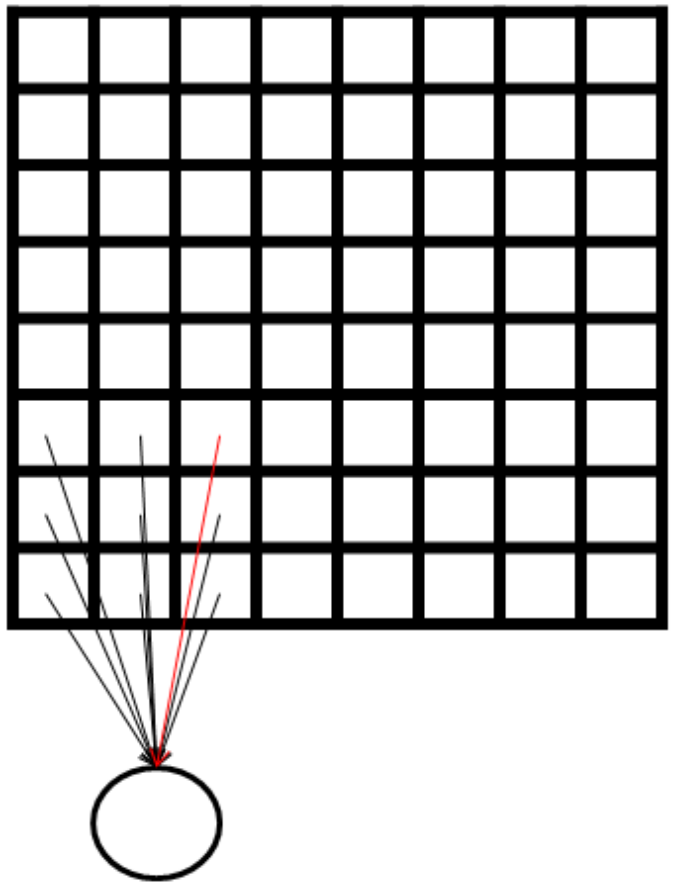
https://www.owlnet.rice.edu/~elec539/Projects97/morphjrks/moredge.html

Edge Detectors, Filters, Features and Maps





A view on what is happening

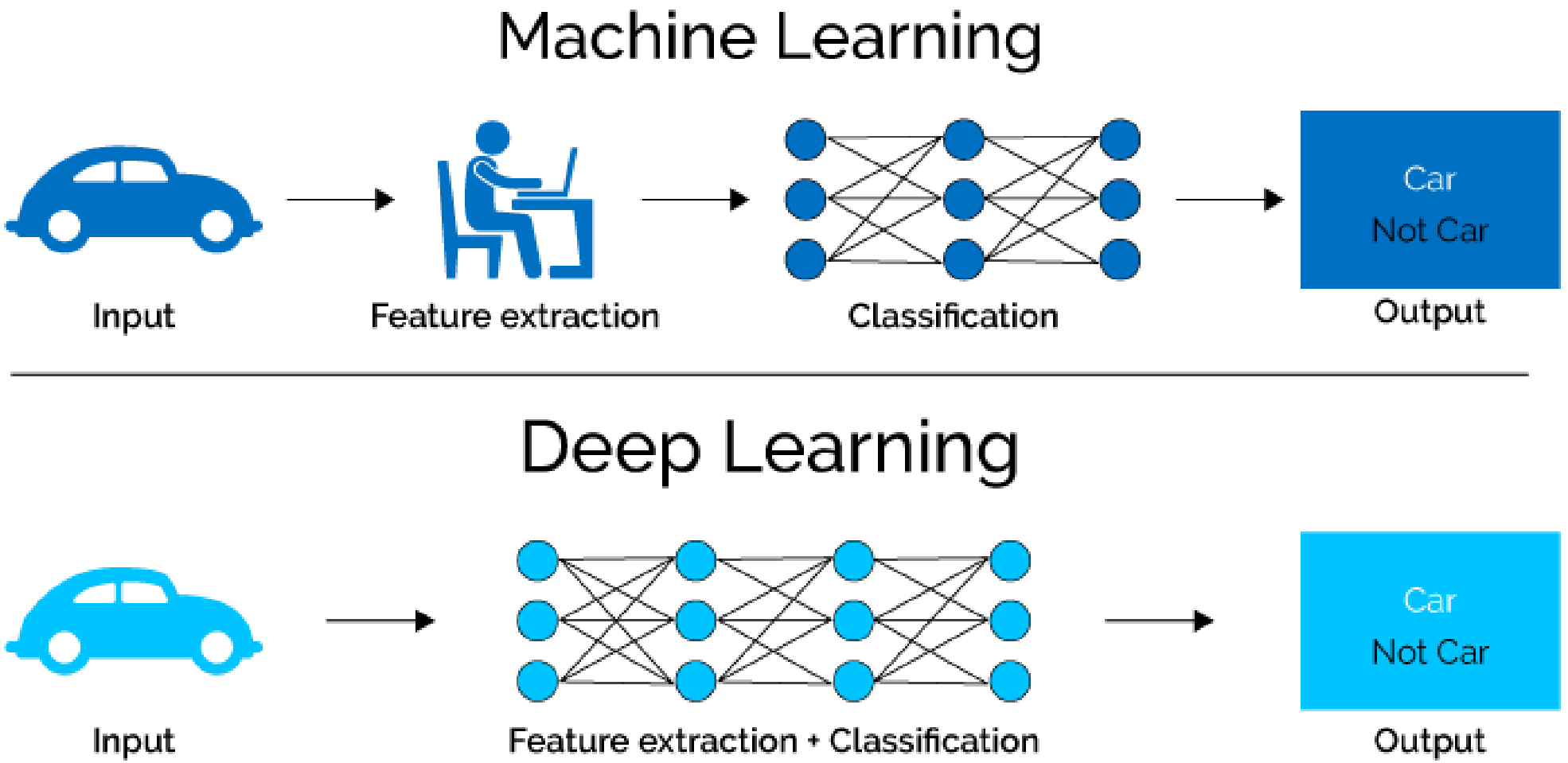


Quick Task

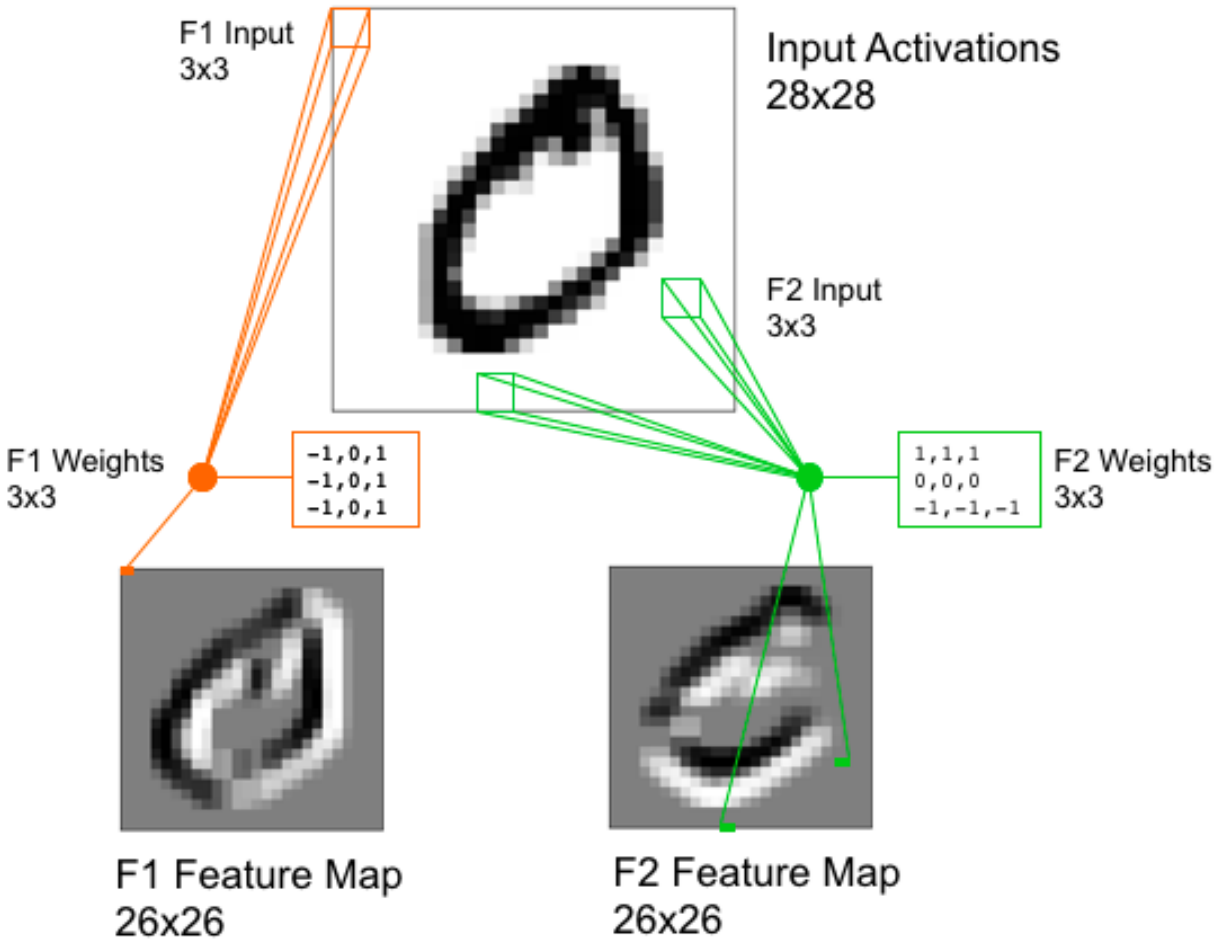
- Re-run my edge detector code for a new input image.
- Re-run the code with vertical-edge filter.

- Lab: Implement a kernel / filter that produces both vertical and horizontal filtering together.

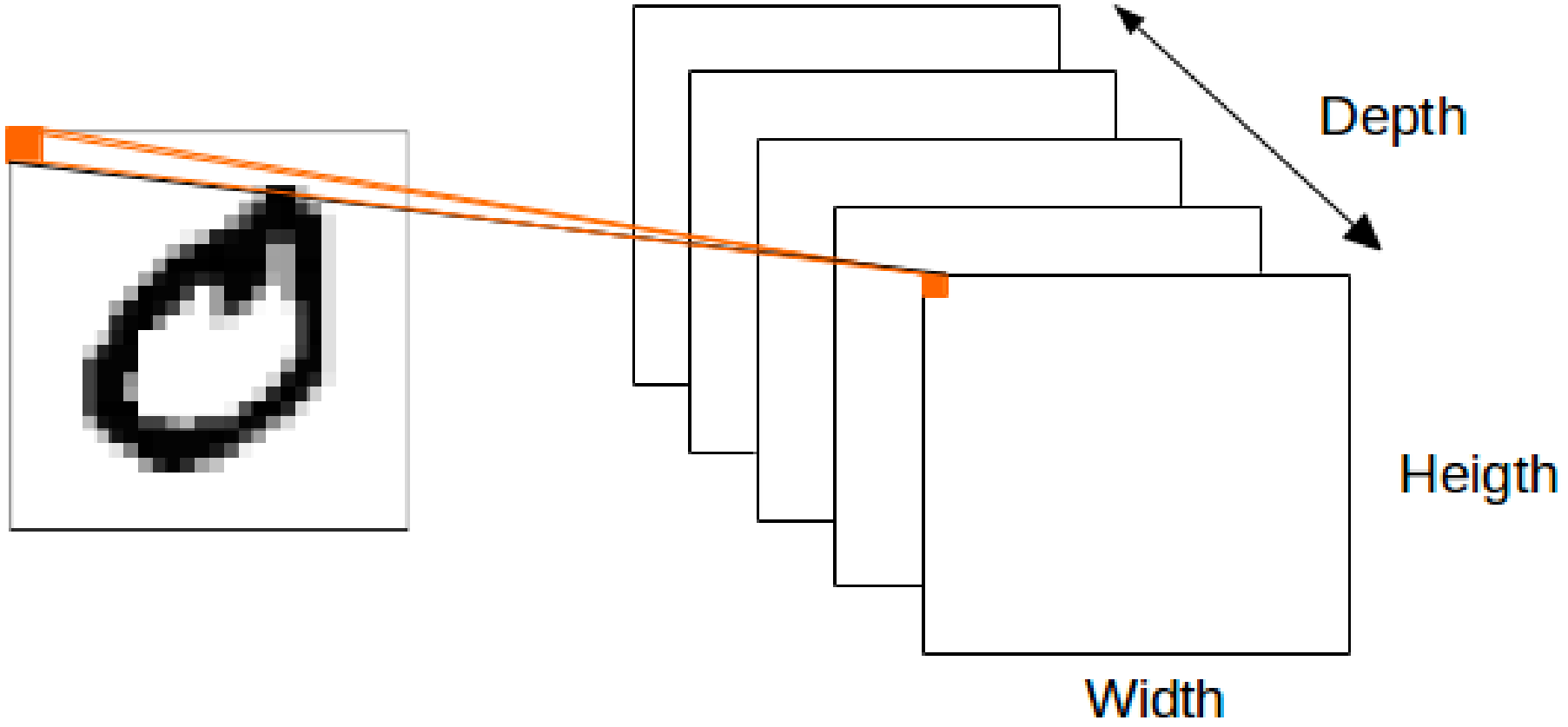
CNNs - Let's Automate This



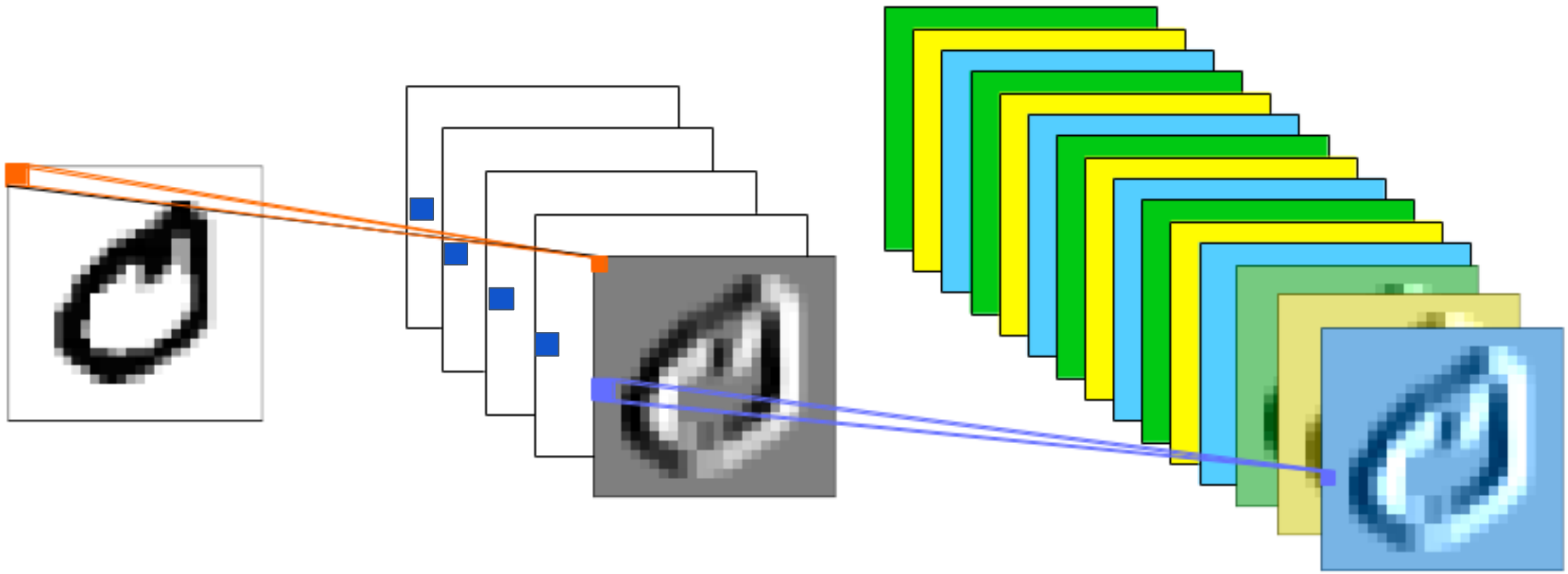
Edge Detectors, Filters, Features and Maps



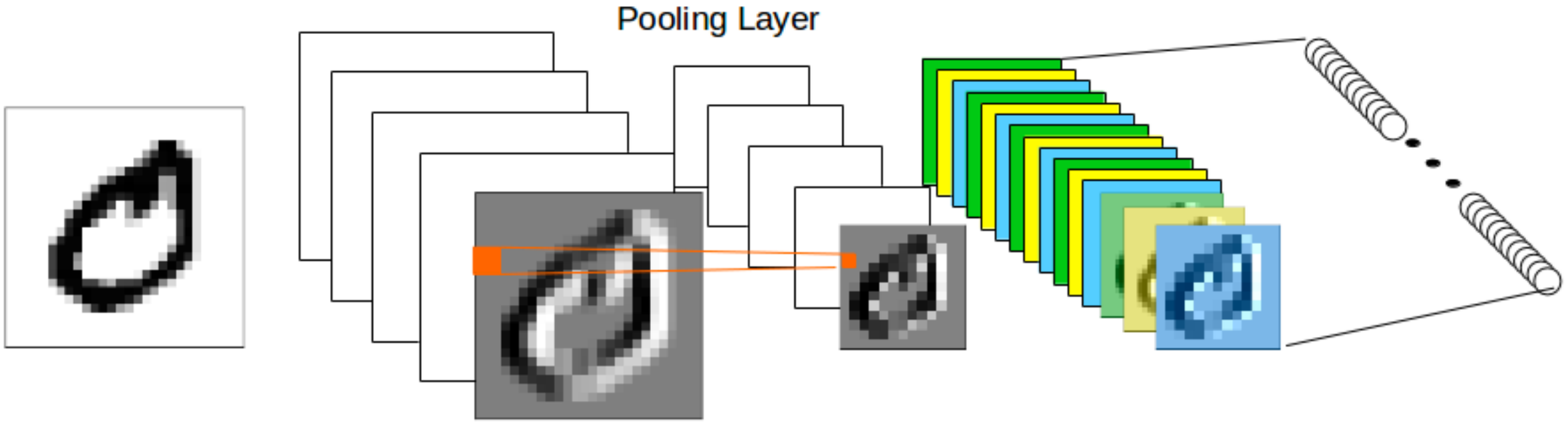
Convolutional Layer Parameters



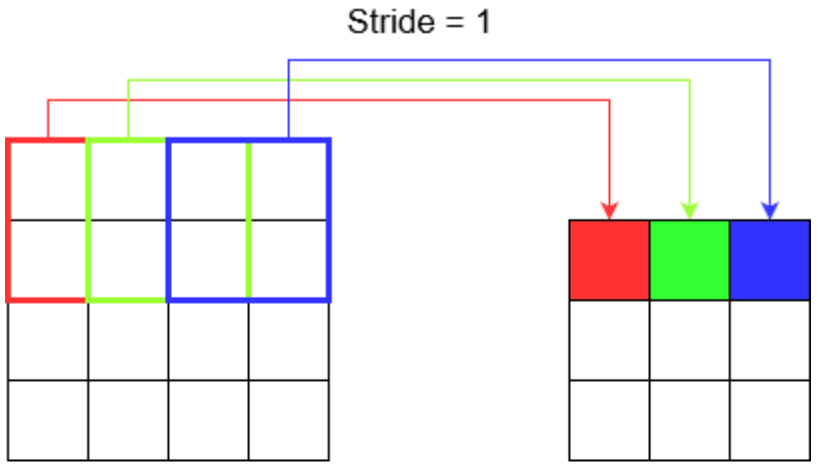
Multiple Convolutional Layers

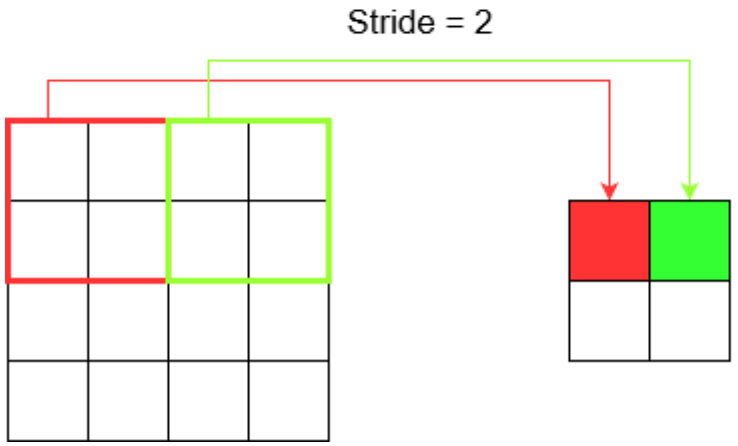


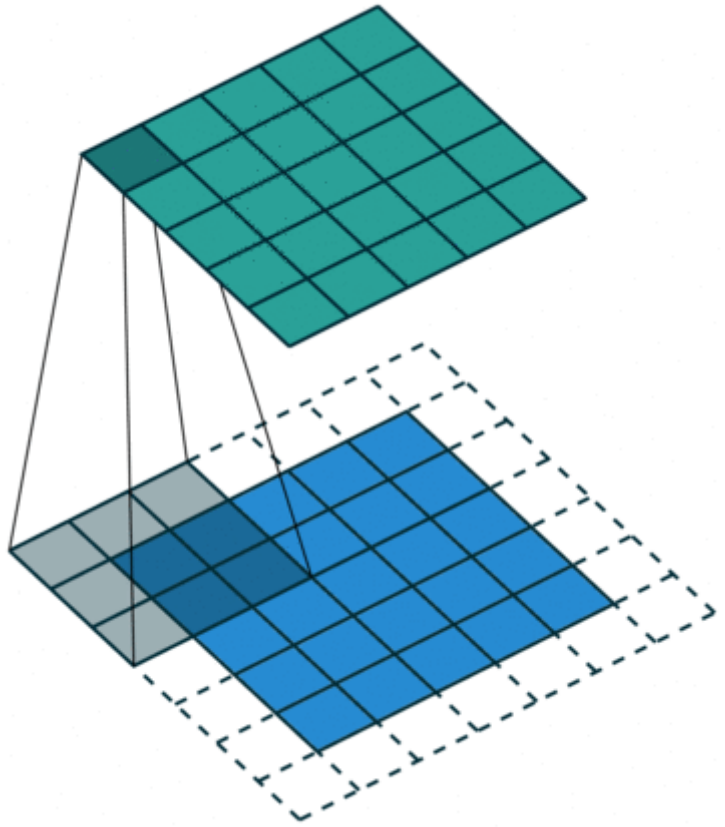
Pooling



Stride and Padding



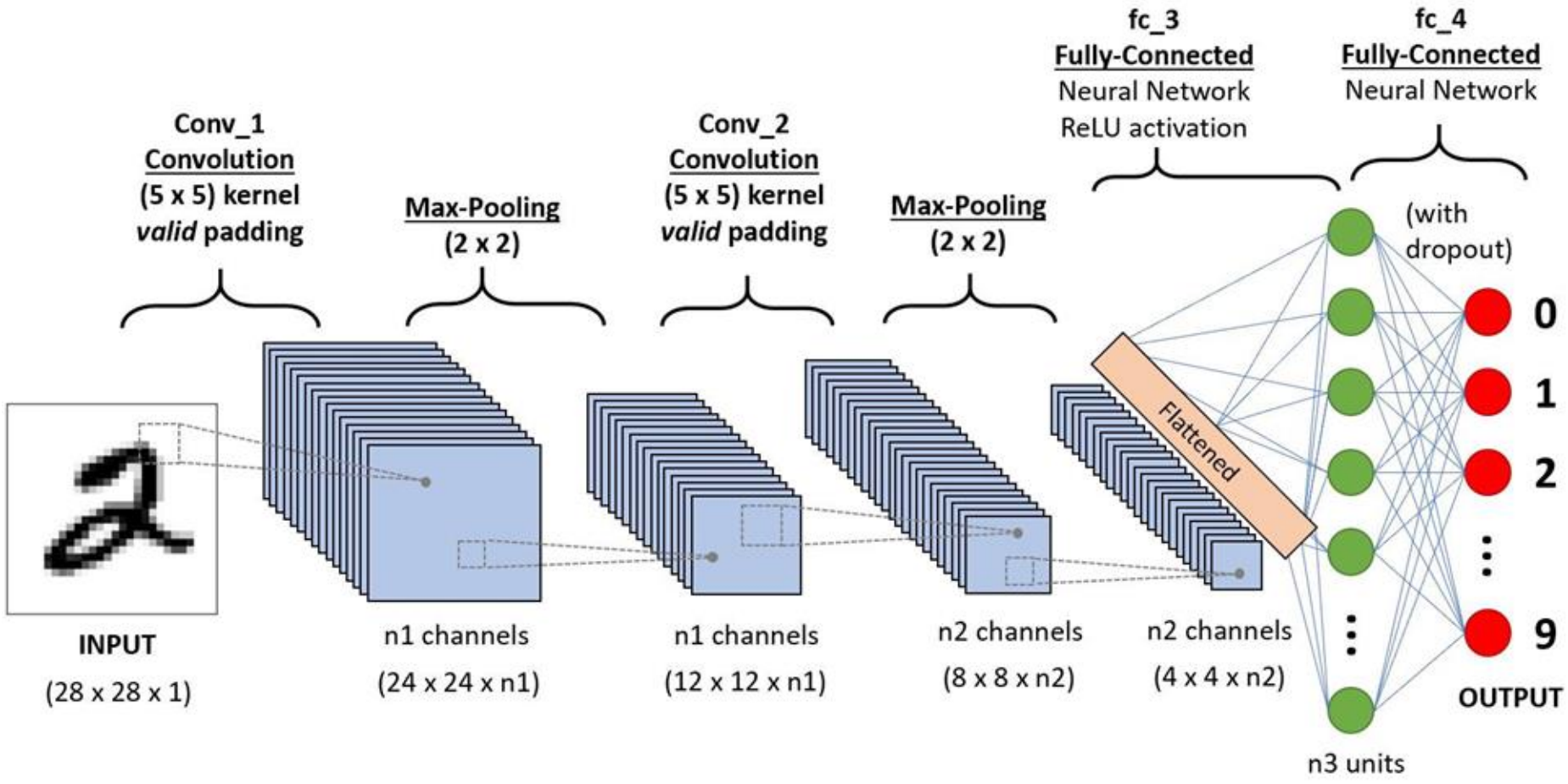




https://towardsdatascience.com/types-of-convolutions-in-deep-learning-717013397f4d

https://stackoverflow.com/questions/42883547/intuitive-understanding-of-1d-2d-and-3d-convolutions-in-convolutional-neural-n

CNNs for Image Processing



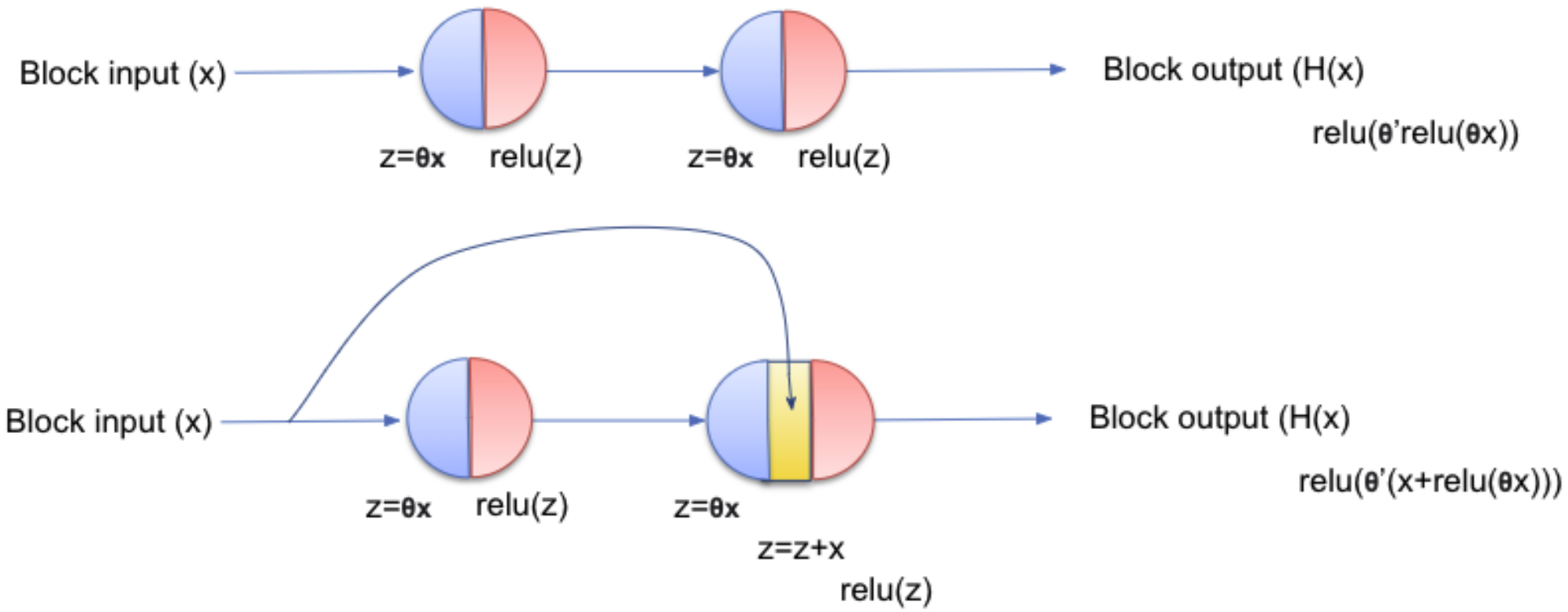
https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53

Group Task

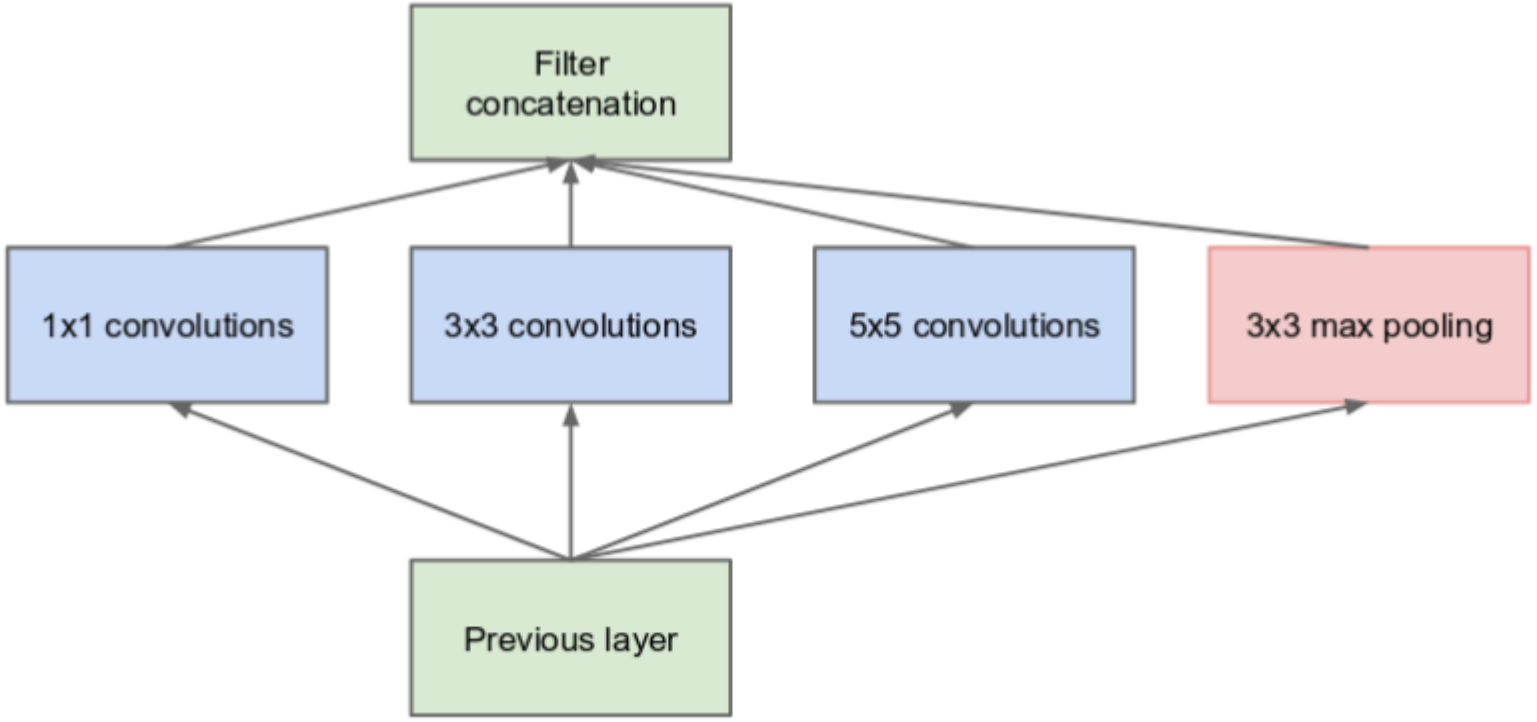
- Experiment with the MNIST example with convolution to try out different parameter variations. As a team discuss the parameter variations that are possible -- make reference to the keras API if necessary for inspiration on functionality variations that are possible.

CNNs to the Next Level

Residual Layers



Inception and other variants



(a) Inception module, naïve version

Inception Extensions



Bharath Raj

May 29, 2018 · 8 min read · Listen

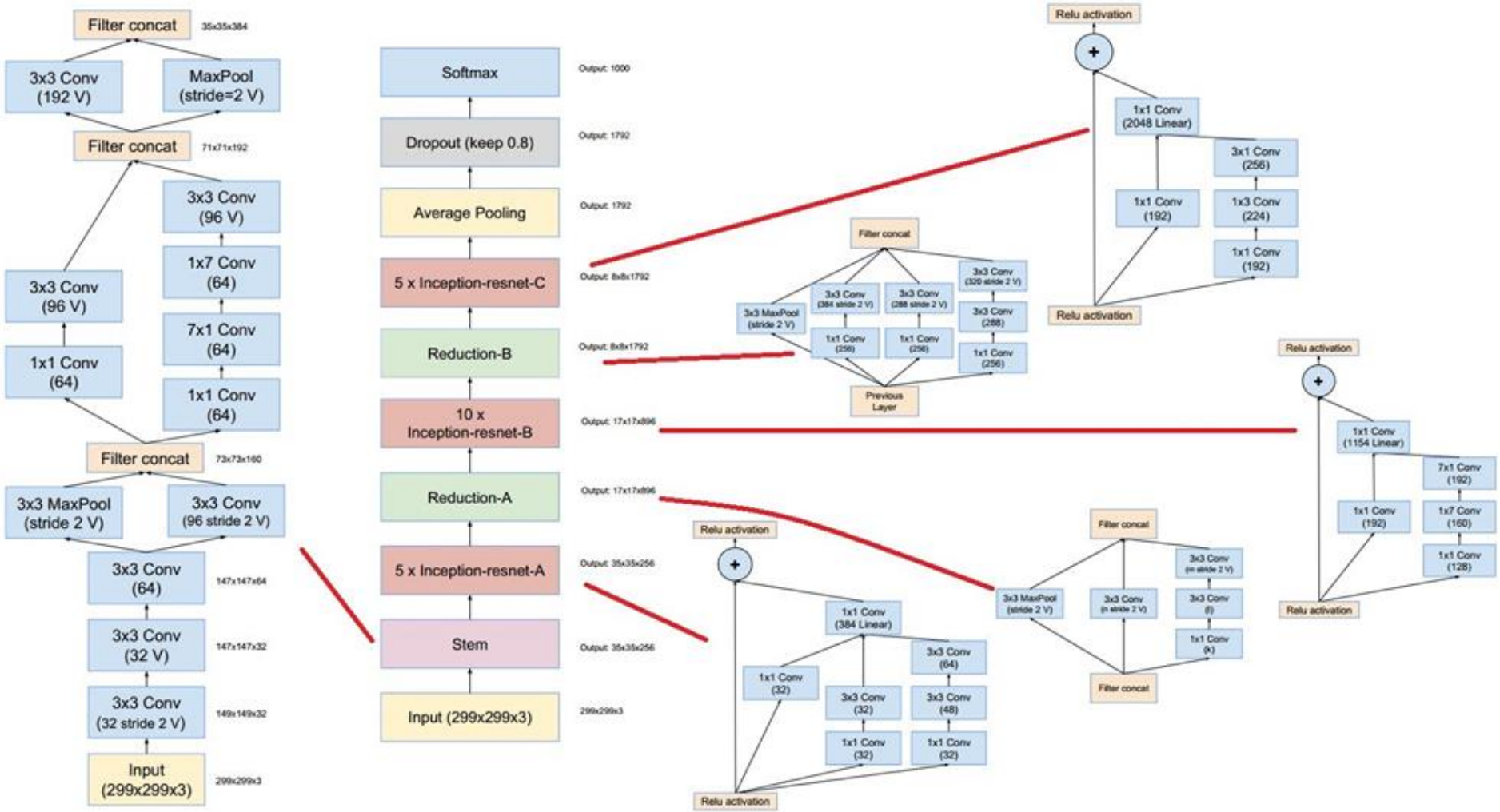
A Simple Guide to the Versions of the Inception Network

The Inception network was an important milestone in the development of CNN classifiers. Prior to its inception (pun intended), most popular CNNs just stacked convolution layers deeper and deeper, hoping to get better performance.

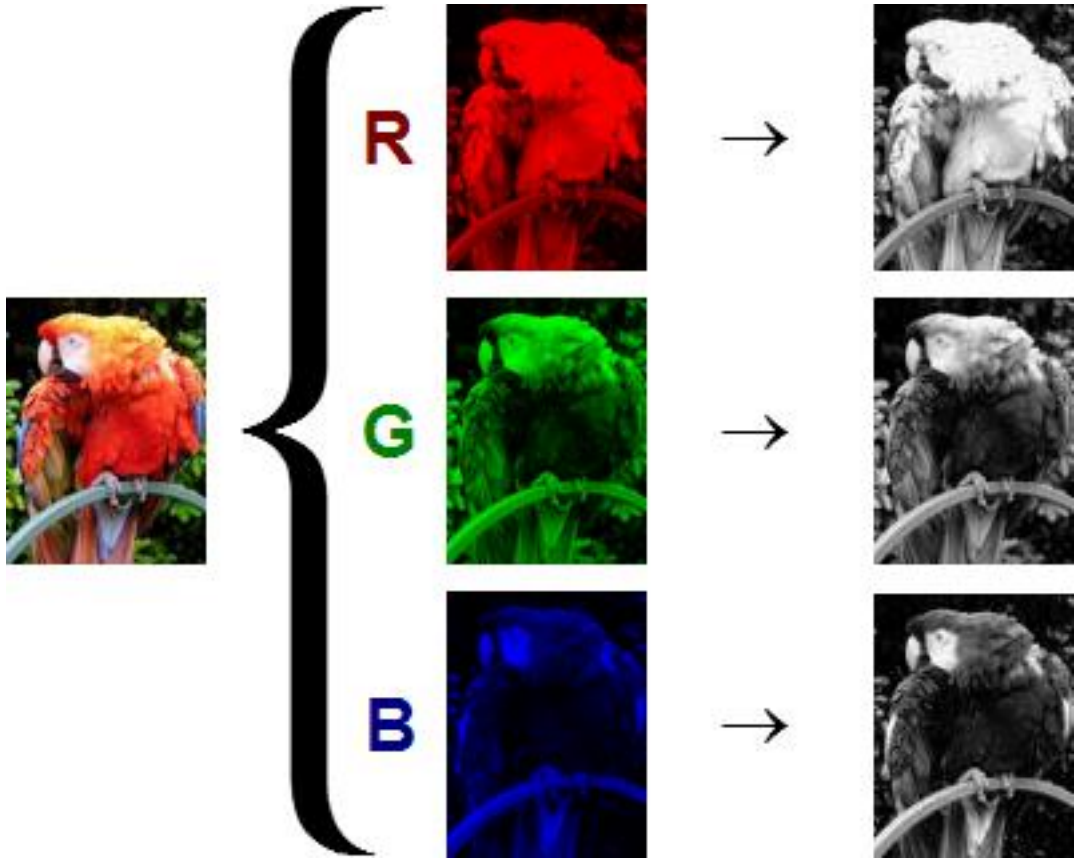


Designing CNNs in a nutshell. Fun fact, this meme was referenced in the first inception net paper.

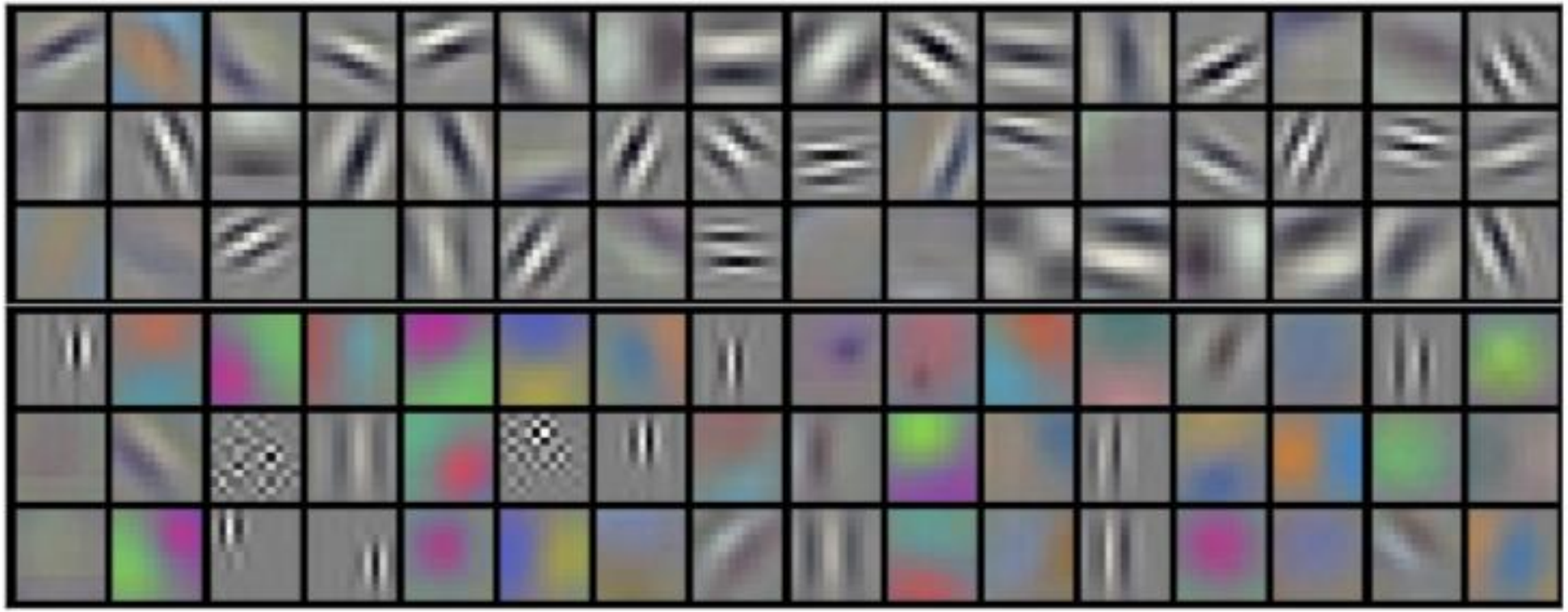
Inception


Filter concat
35x35x384
3x3 Conv (192 V)
MaxPool (stride=2 V)
Filter concat
71x71x192
3x3 Conv (96 V)
1x7 Conv (64)
7x1 Conv (64)
1x1 Conv (64)
3x3 Conv (96 V)
1x1 Conv (64)
Filter concat
73x73x160
3x3 MaxPool (stride 2 V)
3x3 Conv (96 stride 2 V)
3x3 Conv (64)
147x147x64
3x3 Conv (32 V)
147x147x32
3x3 Conv (32 stride 2 V)
149x149x32
Input (299x299x3)
299x299x3
Softmax
Output: 1000
Dropout (keep 0.8)
Output: 1792
Average Pooling
Output: 1792
5 x Inception-resnet-C
Output: 8x8x1792
Reduction-B
Output: 8x8x1792
10 x Inception-resnet-B
Output: 17x17x896
Reduction-A
Output: 17x17x896
5 x Inception-resnet-A
Output: 35x35x256
Stem
Output: 35x35x256
Input (299x299x3)
299x299x3
Relu activation
1x1 Conv (2048 Linear)
3x1 Conv (256)
1x1 Conv (192)
1x3 Conv (224)
1x1 Conv (192)
Relu activation
Filter concat
3x3 MaxPool (stride 2 V)
3x3 Conv (384 stride 2 V)
3x3 Conv (288 stride 2 V)
3x3 Conv (320 stride 2 V)
3x3 Conv (288)
1x1 Conv (256)
1x1 Conv (256)
1x1 Conv (256)
Previous Layer
Relu activation
1x1 Conv (1154 Linear)
7x1 Conv (192)
1x1 Conv (192)
1x7 Conv (160)
1x1 Conv (128)
Relu activation
Relu activation
1x1 Conv (384 Linear)
3x3 Conv (64)
1x1 Conv (32)
3x3 Conv (32)
3x3 Conv (48)
1x1 Conv (32)
1x1 Conv (32)
Relu activation
Filter concat
3x3 MaxPool (stride 2 V)
3x3 Conv (n stride 2 V)
3x3 Conv (n)
1x1 Conv (k)
Filter concat

Basic Image Representation



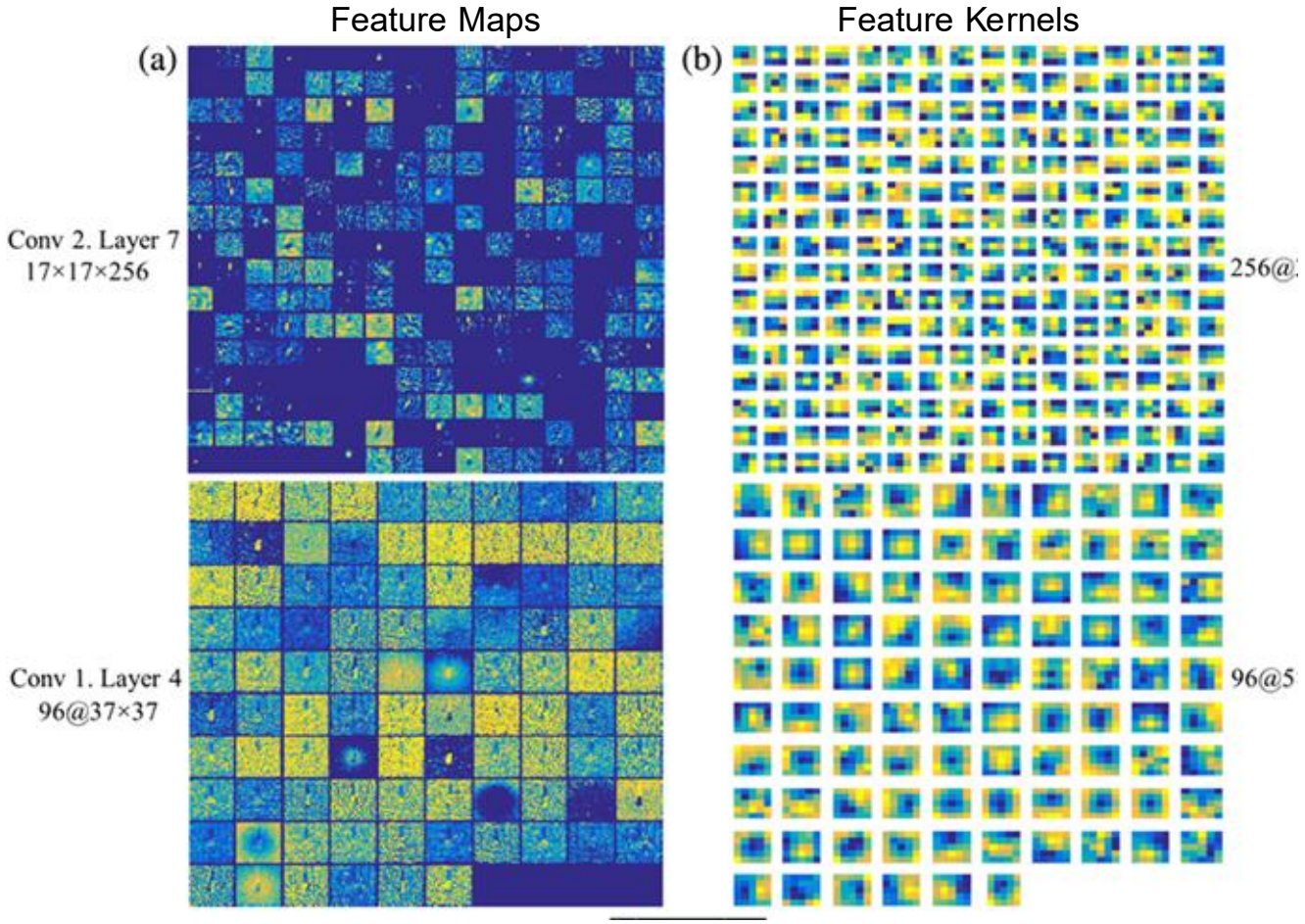
What do Feature Maps Look Like?



Learnings of First convolution layer on image of size 224X224X3

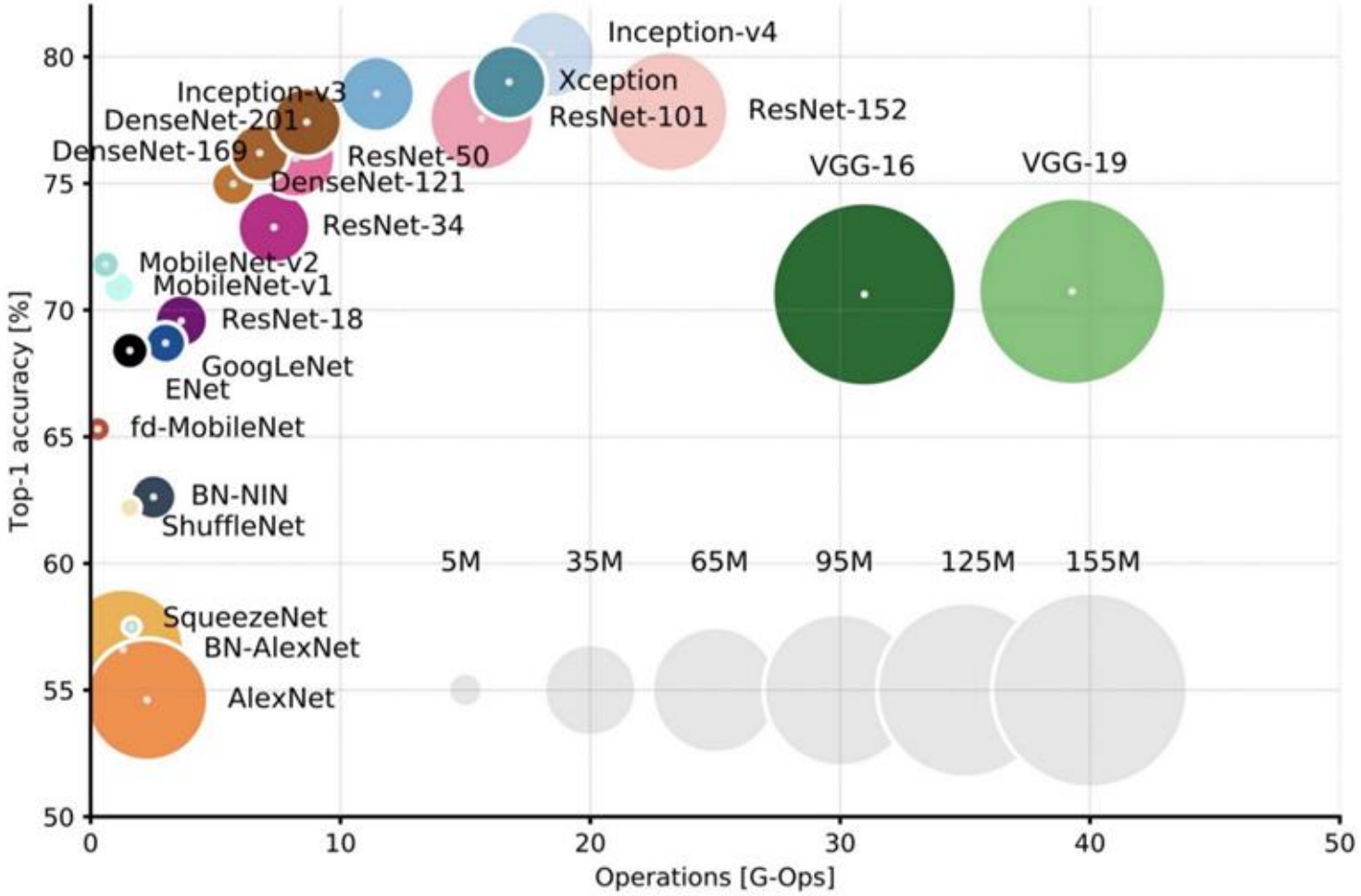
https://software.intel.com/content/www/us/en/develop/articles/visualising-cnn-models-using-pytorch.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+ISNMain+%28Intel+Developer+Zone+Articles+Feed%29

What do Feature Maps Look Like?



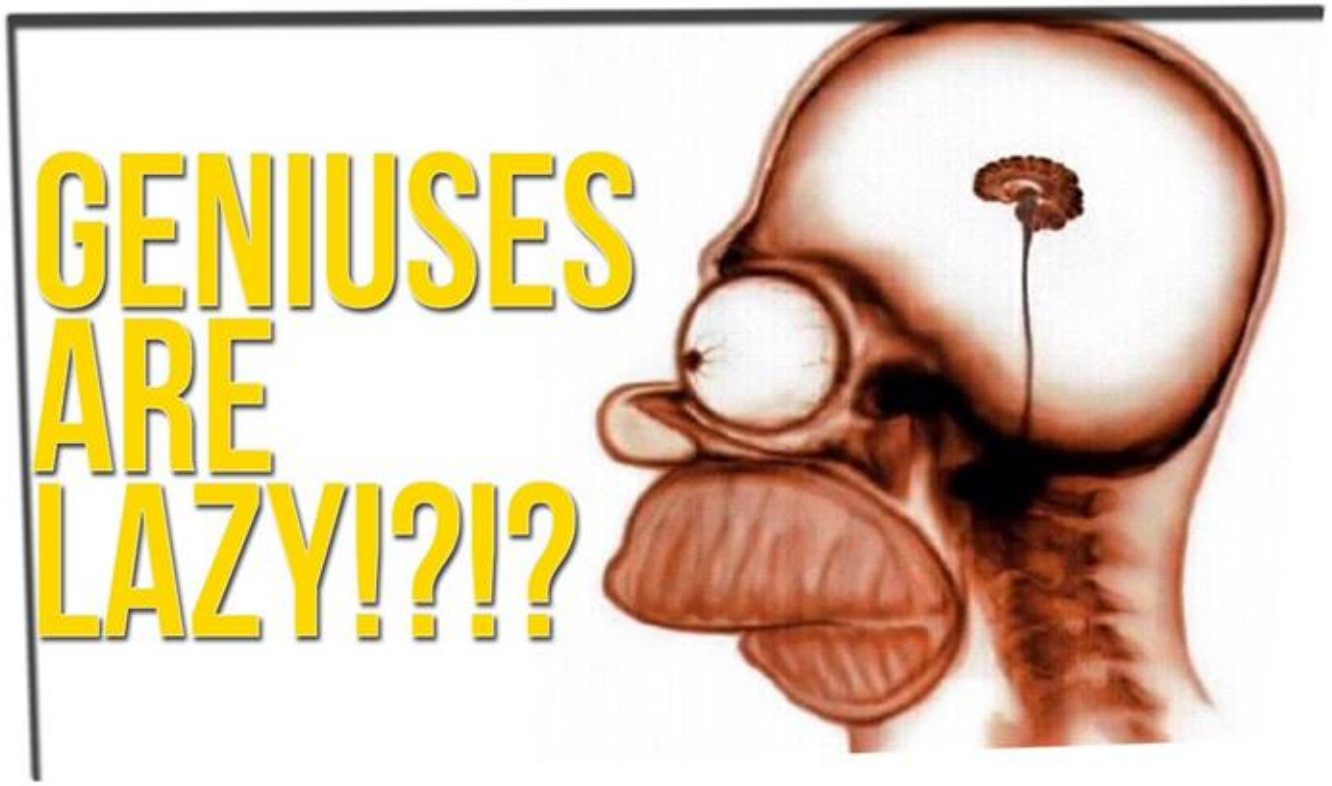
https://www.researchgate.net/publication/320461081_Deep_feature_extraction_and_combination_for_synthetic_aperture_radar_target_classification/figures?lo=1

Comparison of different CNNs



https://towardsdatascience.com/neural-network-architectures

Pre-Trained Weights

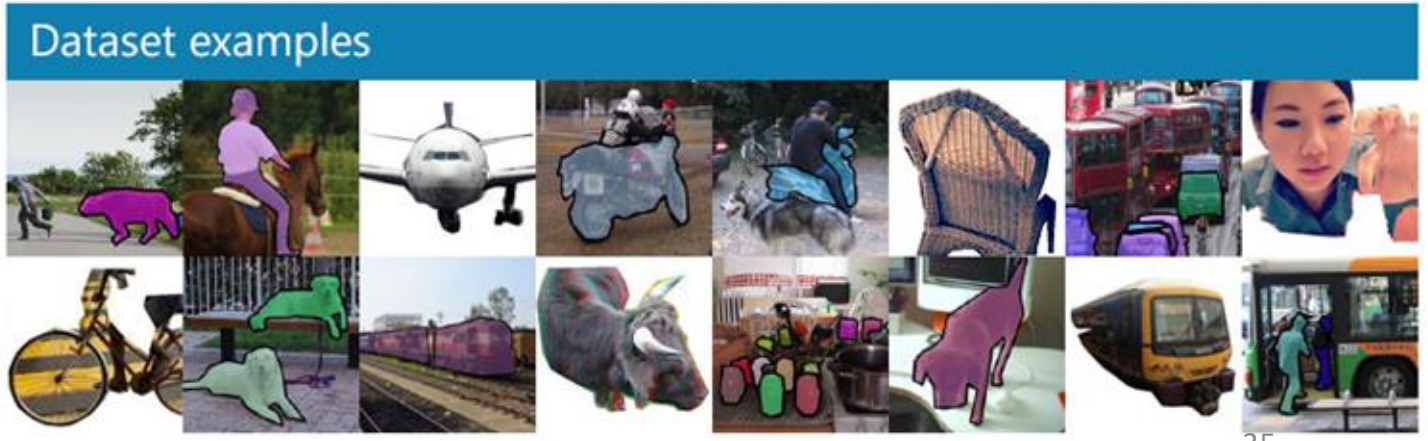


We return to this in the coming weeks…

Sources of Training Data



Imagenet & MS COCO



The end